

A Novel Statistical Recognition Method Based on Hypersphere Model for Radar HRRP ATR

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Abstract

Different from general Gaussian-distributed data, L_2 normalized samples are applied to HRRP-based recognition to deal with the amplitude-scale sensitivity problem, therefore, geometrically speaking, power transformed HRRP samples spread on a unit hypersphere. This paper proposes a novel statistical recognition method for power transformed HRRP samples under the jointly multivariate Gaussian distribution hypothesis, in which the hyperspherical spread of HRRP samples and the effectively discriminating information contained in the noise subspace can be fully utilized without increasing computation complexity. The experimental results based on measured data show that our proposed method can greatly improve the recognition performance.

1. Introduction

A high-resolution range profile (HRRP) is the amplitude of the coherent summations of the complex time returns from target scatterers in each range cell, which represents the projection of the complex returned echoes from the target scattering centers onto the radar line-of-sight (LOS). It contains the target structure signatures, such as target size, scatterer distribution, etc., thereby radar HRRP target recognition has received intensive attention from the radar automatic target recognition (RATR) community[1~4]. By statistical recognition is meant that the class membership of a measurement HRRP x is determined based on the posterior probabilities of all class memberships $P(k|x)$ ($k=1,2,\dots,K$ denotes class membership). How to accurately describe HRRP's statistical characteristics is a challenging task in radar HRRP-based statistical recognition. Some simple statistical models[1,3,4] were proposed for HRRP

samples under the hypothesis that the elements in an HRRP sample are statistically independent. According to the scattering center model, the echo from a range resolution cell varies with the distance differences between the scatterers in this cell. When a target rotates, the scatterers from different range resolution cells will rotate in the same way, which causes that the scatterers' distance differences in different range resolution cells dependently change, therefore, the statistical properties of the echoes from different range resolution cells are also dependent to some extent. Thus a statically correlated model is expected to describe the HRRP's statistical characteristic more accurately. Some literatures[2] showed that the power transformation or the Box-Cox transformation with proper parameters can correct for the departures from Gaussian distribution to some extent. Thus jointly multivariate Gaussian model can be used for statistical recognition based on power transformed HRRP samples.

As we know, the conventional principal components analysis (PCA)-based subspace statistical recognition method can be used for jointly multivariate Gaussian-distributed data. However, Different from general Gaussian-distributed data, L_2 normalized samples are applied to HRRP-based recognition to deal with the amplitude-scale sensitivity problem[4], therefore, geometrically speaking, power transformed HRRP samples spread on a unit hypersphere. This paper will propose a novel statistical recognition method based on hypersphere model for power transformed HRRP samples under the jointly multivariate Gaussian distribution hypothesis.

2. Conventional PCA-based subspace statistical recognition method

Let $X_{kl} = \{x_{kli} \in \mathbb{R}^d | i=1,2,\dots,N\}$ denote the l th ($l=1,2,\dots,L_k$) HRRP frame of target T_k ($k=1,2,\dots,K$), of which each d -dimensional sample x_{kli} (Usually d is

large, e.g. $d = 256$ for our measured data.) are power transformed, range aligned and amplitude normalized. Assume that the HRRP frame follows jointly multivariate Gaussian distribution, namely a sample from this frame $\mathbf{x} \sim \mathcal{N}(\mathbf{m}_{\mathbf{x}|kl}, \mathbf{C}_{\mathbf{x}|kl})$,

$$\mathbf{m}_{\mathbf{x}|kl} \approx \frac{1}{N} \sum_{i=1}^N \mathbf{x}_{kli} \quad (1)$$

$$\mathbf{C}_{\mathbf{x}|kl} \approx \frac{1}{N} \sum_{i=1}^N (\mathbf{x}_{kli} - \mathbf{m}_{\mathbf{x}|kl})(\mathbf{x}_{kli} - \mathbf{m}_{\mathbf{x}|kl})^T = \mathbf{U}_{kl(d)} \mathbf{A}_{kl(d)} \mathbf{U}_{kl(d)}^T \quad (2)$$

where $\mathbf{U}_{kl(d)} = [\mathbf{u}_{kl1} \ \mathbf{u}_{kl2} \ \cdots \ \mathbf{u}_{kl d}]$ and

$\mathbf{A}_{kl(d)} = \text{diag}([\lambda_{kl1} \ \lambda_{kl2} \ \cdots \ \lambda_{kl d}])$ respectively denote the corresponding eigenvector matrix and eigenvalue matrix of $\mathbf{C}_{\mathbf{x}|kl}$ with $\lambda_{kl1} \geq \lambda_{kl2} \geq \cdots \geq \lambda_{kl d}$. The covariance matrix's PCA-based subspace feature vector of $\mathbf{x}$ is

$$\mathbf{y} = \mathbf{U}_{kl(q)}^T (\mathbf{x} - \mathbf{m}_{\mathbf{x}|kl}) = [\mathbf{u}_{kl1} \ \mathbf{u}_{kl2} \ \cdots \ \mathbf{u}_{kl q}]^T (\mathbf{x} - \mathbf{m}_{\mathbf{x}|kl}) \quad (3)$$

where $q < d$. Due to the Gaussian distribution assumption and the orthogonality between $\mathbf{u}_{klj}$ ($j=1,2,\dots,q$), y_j ($j=1,2,\dots,q$) should be independent and follow Gaussian distribution, namely $\mathbf{y} \sim \mathcal{N}(\mathbf{m}_{\mathbf{y}|kl}, \mathbf{C}_{\mathbf{y}|kl})$ with

$$\mathbf{m}_{\mathbf{y}|kl} \approx \mathbf{U}_{kl(q)}^T \left(\frac{1}{N} \sum_{i=1}^N \mathbf{x}_{kli} - \mathbf{m}_{\mathbf{x}|kl} \right) \approx \mathbf{0}_{(q)} \quad (4)$$

$$\mathbf{C}_{\mathbf{y}|kl} \approx \mathbf{U}_{kl(q)}^T \mathbf{C}_{\mathbf{x}|kl} \mathbf{U}_{kl(q)} = \mathbf{U}_{kl(q)}^T (\mathbf{U}_{kl(d)} \mathbf{A}_{kl(d)} \mathbf{U}_{kl(d)}^T) \mathbf{U}_{kl(q)} \quad (5)$$

$$= \mathbf{A}_{kl(q)}$$

where $\mathbf{0}_{(q)}$ denoting a $q \times 1$ vector with zero elements.

Thus the statistical model for the covariance matrix's PCA-based subspace of each HRRP frame can be established. Obviously, this method utilizes the statistical characteristics of the PCA-based subspace in the covariance matrix, thus is referred to as the PCA-cov subspace method in this paper.

3. A novel statistical recognition method

3.1. Hyperspherical spread of HRRP samples

As discussed in Section I, different from general Gaussian-distributed data, amplitude L_2 normalized HRRP samples are applied to RATR, in other words, geometrically speaking, HRRP samples in RATR spread on a unit hypersphere. The correlation matrix of $\mathbf{X}_{kl} = \{\mathbf{x}_{kli} \in \mathbb{R}^d \mid i=1,2,\dots,N\}$ is

$$\mathbf{R}_{\mathbf{x}|kl} = E[\mathbf{x}\mathbf{x}^T] \approx \frac{1}{N} \sum_{i=1}^N \mathbf{x}_{kli} \mathbf{x}_{kli}^T = \mathbf{V}_{kl(d)} \mathbf{\Sigma}_{kl(d)} \mathbf{V}_{kl(d)}^T \quad (6)$$

where $\mathbf{V}_{kl(d)} = [\mathbf{v}_{kl1} \ \mathbf{v}_{kl2} \ \cdots \ \mathbf{v}_{kl d}]$ and

$\mathbf{\Sigma}_{kl(d)} = \text{diag}([\varsigma_{kl1} \ \varsigma_{kl2} \ \cdots \ \varsigma_{kl d}])$ respectively denote the corresponding eigenvector matrix and eigenvalue matrix of $\mathbf{R}_{\mathbf{x}|kl}$ with $\varsigma_{kl1} \geq \varsigma_{kl2} \geq \cdots \geq \varsigma_{kl d}$. Mathematically speaking, $\mathbf{v}_{kl1}$, the first eigenvector of $\mathbf{R}_{\mathbf{x}|kl}$, satisfied

$$\begin{cases} \mathbf{v}_{kl1} = \arg \min_{\mathbf{v}} \sum_{i=1}^N \|\mathbf{x}_{kli} - \langle \mathbf{x}_{kli}, \mathbf{v} \rangle \mathbf{v}\|_2^2 \\ \text{s.t. } \|\mathbf{v}\|_2 = 1 \end{cases} \quad (7)$$

where $\|\cdot\|_2$ denotes the L_2 norm operation of a vector, $\langle \cdot, \cdot \rangle$ denotes the inner product operation between two vectors; while the average vector of this data set satisfies

$$\mathbf{m}_{\mathbf{x}|kl} = \arg \min_{\boldsymbol{\mu}} \sum_{i=1}^N \|\mathbf{x}_{kli} - \boldsymbol{\mu}\|_2^2 \quad (8)$$

Therefore, for a given HRRP frame $\mathbf{X}_{kl}$, its $\mathbf{m}_{\mathbf{x}|kl}$ and $\mathbf{v}_{kl1}$ should be different. Nevertheless, we can deduce that the average vector and the first eigenvector of the correlation matrix of a data set will be similar, if and only if the data set compactly assembles on the hypersphere. Under the limit condition that the data set assembles so compactly that the hypersphere which the data set assembles on turns into the tangent hyperplane, the average vector and the first eigenvector of the correlation matrix will be exactly same. Figs. 1 (a) and (b) compare $\mathbf{m}_{\mathbf{x}}$, the average vector, and $\mathbf{v}_1$, the first eigenvector of the correlation matrix, of a power transformed HRRP frame with $\alpha=0.7$ in the measured data of "Yark-42", which show the waveforms of the two vectors are similar. Therefore, an HRRP frame compactly assembles on the unit hypersphere. Moreover, Figs. 1 (c) and (d) show that the waveforms of $\mathbf{u}_1$, the first eigenvector of the covariance matrix, and $\mathbf{v}_2$, the second eigenvector of the correlation matrix, of this frame are also similar with $\lambda_1 \approx \varsigma_2$. Similarly, experimental results show $\mathbf{u}_2 \approx \mathbf{v}_3$, $\lambda_2 \approx \varsigma_3$ etc., and only the small eigenvalues and the corresponding eigenvectors have evident difference. Therefore, we can draw the conclusion that the average vector and the correlation matrix's first eigenvector of an HRRP frame approximate to be in the same direction, and the PCA-based subspace in the covariance matrix approximates to be same as the subspace made up of other eigenvectors with larger eigenvalues in the correlation matrix from the experimental results.

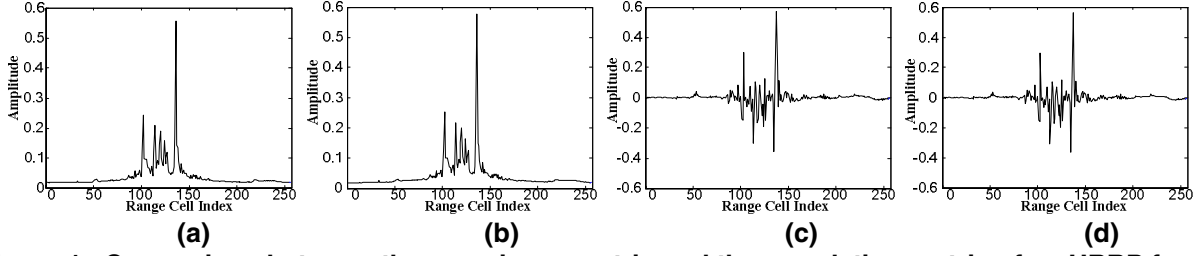


Figure 1. Comparison between the covariance matrix and the correlation matrix of an HRRP frame with power parameter $\alpha = 0.7$ of "Yark-42": (a) the average vector m_x ; (b) the first eigenvector of the correlation matrix v_1 ; (c) the first eigenvector of the covariance matrix u_1 with $\lambda_1 = 0.0161$; (d) the second eigenvector of the correlation matrix v_2 with $\lambda_2 = 0.0161$.

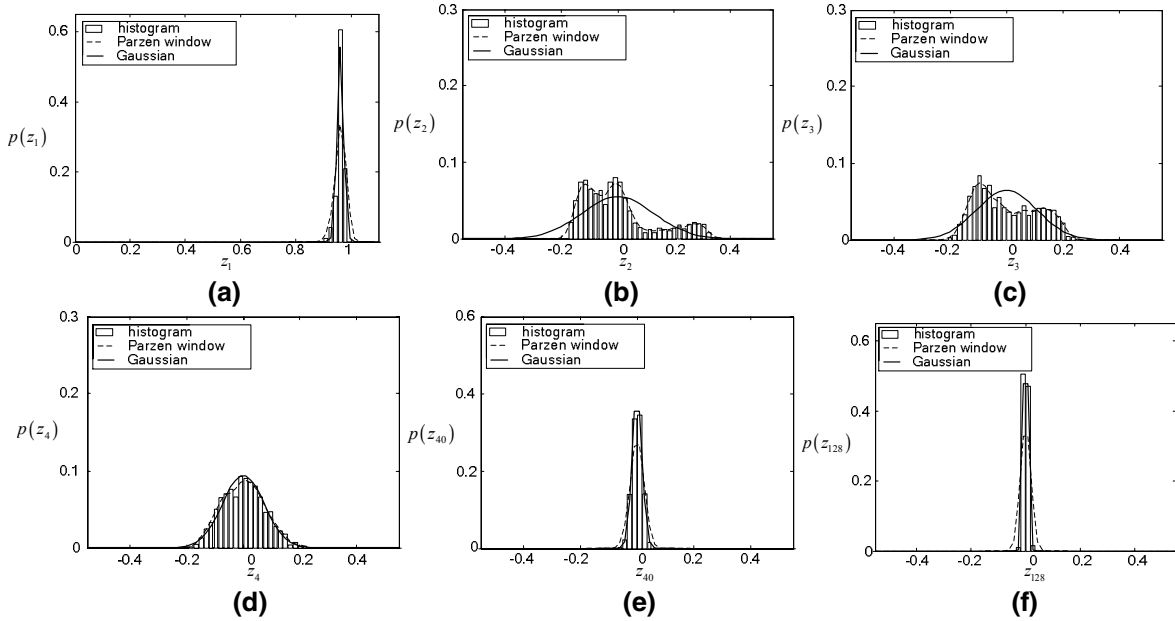


Figure 4. The approximated pdfs of some elements in the correlation matrix's PCA-based subspace feature vector from the HRRP frame same as that used in Figure 3:

(a) z_1 with $m_{z_1} = 0.9601$ and $C_{z_1} = 1.57 \times 10^{-4}$; (b) z_2 ; (c) z_3 ; (d) z_4 ; (e) z_{40} ; (f) z_{128} .

Similar to Eq. (3), the correlation matrix's PCA-based subspace feature vector of x is defined as

$$z = V_{kl(q)}^T x = [v_{kl1} \ v_{kl2} \ \cdots \ v_{klq}]^T x \quad (9)$$

Due to the Gaussian distribution assumption and the orthogonality between v_{klj} ($j = 1, 2, \dots, q$), z_j ($j = 1, 2, \dots, q$) should be independent and follow Gaussian distribution, namely $z \sim \mathcal{N}(m_{z|kl}, C_{z|kl})$, by formula derivation,

$$m_{z|kl} \approx \frac{1}{N} \sum_{i=1}^N V_{kl(q)}^T x_{kli} = V_{kl(q)}^T \left(\frac{1}{N} \sum_{i=1}^N x_{kli} \right) \approx V_{kl(q)}^T m_{x|kl} \quad (10)$$

$$C_{z|kl} \approx \frac{1}{N} \sum_{i=1}^N V_{kl(q)}^T (x_{kli} - m_{x|kl})(x_{kli} - m_{x|kl})^T V_{kl(q)} \quad (11)$$

$$\approx V_{kl(q)}^T C_{x|kl} V_{kl(q)}$$

In the following, we will make a detailed analysis on the statistical characteristic of z_j ($j = 1, 2, \dots, q$), the

element in the correlation matrix's PCA-based subspace feature vector, based on measured data. Figs. 3 (a)~(f) show the approximated probability density functions (pdfs) of some elements in the correlation matrix's PCA-based subspace feature vectors from the HRRP frame same as that used in Fig. 2, where the bar represents that approximated by histogram method, the dotted line represents that by Parzen windows using Gaussian kernels. According to the aforementioned analysis on hyperspherical spread of HRRP samples, the projection of each HRRP sample in a frame to the first eigenvector of the correlation matrix will be a little smaller than 1, and the mean of the projections of all HRRP samples in a frame to another eigenvector of the correlation matrix will approximate to 0, therefore, in theory, z_1 should follow a narrow distribution with nonzero mean, where $m_{z_1} \approx v_1^T m_x$ approximates to 1

$C_{z_j} \approx \mathbf{v}_j^T \mathbf{C}_x \mathbf{v}_j = \mathbf{v}_j^T (\mathbf{U}_d \mathbf{A}_d \mathbf{U}_d^T) \mathbf{v}_j \approx \lambda_{j-1} \approx \varsigma_j$. To simplify the statistical model, we assume that all of the elements in the feature vector are independent, and each element follows the corresponding Gaussian distribution. According to the experimental results shown in Figs. 2 (a)~(f), this is true for z_j ($j=1, 4, 5, \dots, q$), of which the solid line represents the corresponding Gaussian pdfs, while z_j ($j=2, 3$) follows multimodal distribution. This is caused by the fact that the power transformed HRRP samples do not follow jointly multivariate Gaussian distribution strictly. Nevertheless, our experimental results show that the feature elements evidently departing from Gaussian distribution (usually 2~3) from power transformed HRRP samples are much less than those from original HRRP samples (usually 20~30). This demonstrates that the power transformation with proper power parameters can correct for the departures from Gaussian distribution to some extent. Since after proper power transformation the feature elements not following Gaussian distribution are a few, we still assume that $z_j \sim \mathcal{N}(0, \varsigma_j)$, $j=2, 3, \dots, q$. Therefore,

$\mathbf{C}_{z|kl} \approx \begin{bmatrix} \mathbf{C}_{z_1|kl} \\ \boldsymbol{\Sigma}_{kl(2 \sim q)} \end{bmatrix}$. Thus the statistical model for the

From the aforementioned analysis, we can see that the PCA-cor subspace is similar to the PCA-cov subspace except for the first eigenvector of the correlation matrix. Actually, the distribution characteristics along this direction should be important in radar HRRP statistical recognition. Under the limit condition that $\mathbf{m}_{x|kl} = \mathbf{v}_{kl}$, $\mathbf{v}_{klj} = \mathbf{u}_{klj-1}$, $\varsigma_{klj} = \lambda_{q|j-1}$ ($j = 2, 3, \dots, q$), by formula derivation we get that $z_1 \sim \delta(z_1 - 1)$ with $\delta(t) = \begin{cases} 1 & t = 0 \\ 0 & t \neq 0 \end{cases}$, $z_j \sim \mathcal{N}(0, \varsigma_{klj})$, $j = 2, 3, \dots, q$, in other words, one-dimensional feature

Similar to Eqs. (3) and (9), the correlation matrix's whole space feature vector of $\mathbf{x}$ is defined as

Due to the Gaussian distribution assumption and the orthogonality between $\mathbf{v}_{k,j}$ ($j=1,2,\dots,d$), $\mathbf{z}_j$ ($j=1,2,\dots,d$) should be independent and follow Gaussian distribution. Obviously, $\mathbf{z}_j = \mathbf{z}_j$ ($j=1,2,\dots,q$), $\mathbf{z}_1 \sim \mathcal{N}(\mathbf{m}_{z_1}, \mathbf{C}_{z_1})$, $\mathbf{z}_{j'} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\zeta}_{j'})$,

$j' = 2, 3, \dots, q$. Let $\bar{\varsigma}_{kl} = \frac{1}{(d-q)} \sum_{i=q+1}^d \varsigma_{klj}$. Then we assume

that $\frac{z}{j''} \sim \mathcal{N}(0, \bar{\zeta}_{kl})$, $j'' = q+1, q+2, \dots, d$. Therefore,

$$p(\mathbf{z} | l; k) = p(\mathbf{z} | l; k) (2\pi\bar{\varsigma}_{kl})^{-\frac{d-q}{2}} \exp \left\{ -\frac{1 - \sum_{j=1}^q (\mathbf{v}_{klj}^T \mathbf{z})^2}{2\bar{\varsigma}_{kl}} \right\} \quad (15)$$

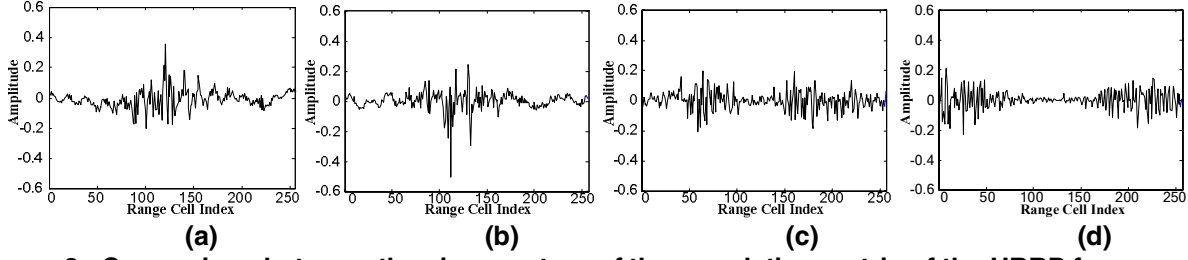


Figure 3. Comparison between the eigenvectors of the correlation matrix of the HRRP frame same as that used in Figures 3 and 4: (a) v_{10} with $\varsigma_{10} = 0.0014$; (b) v_{40} with $\varsigma_{40} = 2.97 \times 10^{-4}$; (c) v_{128} with $\varsigma_{128} = 6.16 \times 10^{-5}$; (d) v_{200} with $\varsigma_{200} = 2.36 \times 10^{-5}$.

Although the aforementioned statistical model utilizes the statistical characteristics of the noise subspace, the computation complexity does not increase, due to the amplitude normalization, i.e. $\sum_{j=1}^d (v_{kj}^T \mathbf{x})^2 = 1$. Compared

with the PCA-cov subspace method and the PCA-cor subspace method, this method utilizes the statistical characteristics of the PCA-based whole space in the correlation matrix, thus is referred to as the PCA-cor space method in this paper.

4. Experimental results

The results presented in this section are based on 3 airplanes' real data, which are measured by radar with the center frequency of 5520 MHz and the bandwidth of 400 MHz. Training data cover almost all of the target-aspect angles of test data, but their elevation angles are different.

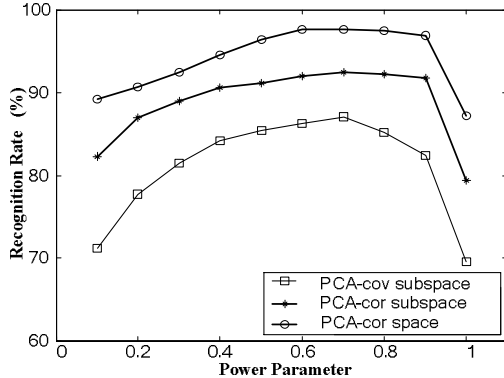


Figure 4. The average recognition rates of the PCA-cov subspace method, the PCA-cor subspace method and the PCA-cor space method versus the power parameter ($q = 40$)

The average recognition rates of the PCA-cov subspace method, the PCA-cor subspace method and the PCA-cor space method versus the power parameter α with $q = 40$ are shown in Fig. 4. For a given power parameter and dimensionality parameter, the average recognition rate of the PCA-cor subspace

method is approximately 5~11 percentage points larger than that of the PCA-cov subspace method, and that of the PCA-cor space method is further larger than that of the PCA-cov subspace method by approximately 4~8 percentage points without increasing computation complexity.

5. Conclusion

A novel statistical recognition method based on the PCA-based subspace in the correlation matrix of each HRRP frame is proposed for power transformed HRRP samples under the jointly multivariate Gaussian distribution hypothesis. Compared with the PCA-cov subspace statistical recognition method, the hyperspherical spread of HRRP samples and the effectively discriminating information contained in the noise subspace can be fully utilized in this method without increasing computation complexity. The experimental results based on measured data show that our proposed method can greatly improve the recognition performance.

6. References

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